Rice dataset feature transition

Raw sources to training-ready CSVs

South Indian rice observations
District, crop year and season

Government crop statistics + GADM locations + NASA weather

Dataset preparation stages

RAW SOURCES

Crop statistics
District boundaries
Daily weather

SELECT

Rice records
Matching fields
Weather channels

MERGE

District identity
Year and season
Seasonal summaries

PREPROCESS

Scale and encode; optionally learn DAE features

EXPORT

Standard and DAE train / validation / test CSVs

Downloaded and assembled raw sources

Source	Local files	Record level
Government crop statistics	crop_statistics.csv	District, crop, year, season
GADM 4.1 district boundaries	india_districts.csv boundary_coordinates.csv	District properties Ordered boundary vertices
NASA POWER daily extract	daily_weather.csv	District and date

GADM originally arrived as zipped GeoJSON. The active folder now contains CSVs preserving all 676 districts and 275,432 vertices.

Three source families feed one seasonal record

CROP STATISTICS

Area and production
District, year, season

BOUNDARIES

District names
Longitude and latitude

DAILY WEATHER

District ID and date
10 weather channels

MATCH AND AGGREGATE

Match district names. Link weather dates to crop year and season. Summarize complete daily records.

One row per district, crop year and season

Crop-statistics feature selection

Raw fields	Selected or calculated fields	Reason
State, District	STATE, DISTRICT, IDREGION	Regional context and district matching
Crop	CROP = Rice	Filter rice / paddy records
Crop_Year, Season	FYEAR, SEASON	Weather alignment and time splits
Area	CROP_AREA, AREA_FRACTION, CROP_AREA_LOG1P	Cultivation area and relative scale
Production + Area	PRODUCTION_TONNES, YIELD	Calculate yield = production / area

Keep valid records from four states. Recompute yield rather than copying raw Yield. Production stays outside predictor inputs to prevent leakage.

District features and matching fields

Boundary information	Use in the dataset
NAME_1 and NAME_2	Match state and district names
Longitude / latitude vertices	Average vertices to obtain CENTROID_X and CENTROID_Y
GID_2	Link district properties to boundary_coordinates.csv
Other GADM codes, names and types	Retain in the district source CSV for traceability

The merge uses representative longitude and latitude. These averages reproduce the original pipeline and are not area-weighted geographic centroids.

Daily weather features become seasonal means

Daily field	Final feature	Meaning
TMAX	TMAX_MEAN	Average daily maximum temperature
TMIN	TMIN_MEAN	Average daily minimum temperature
TAVG	TAVG_MEAN	Average daily mean temperature
VPRES	VPRES_MEAN	Average vapour pressure
WSPD	WSPD_MEAN	Average wind speed
RELH	RELH_MEAN	Average relative humidity
PS	PS_MEAN	Average surface pressure

IDREGION identifies the district. DATE determines which season receives each daily record.

Weather totals and coverage

Daily information	Final feature	Meaning
PREC	PREC_TOTAL	Seasonal rainfall total in mm
ET0	ET0_TOTAL	Seasonal NASA EVLAND land evaporation in mm
RAD	RAD_TOTAL	Seasonal solar radiation in kWh/m²
Complete daily records	WEATHER_DAY_COUNT	Number of days used in the summary

Use a day only when all ten weather channels are finite. Require at least 60 complete days per crop observation. IDCOVER is excluded from the final predictors.

District matching across the sources

GOVERNMENT RECORD

Andhra Pradesh
ADILABAD, 2001, Kharif

GADM RECORD

Telangana
Adilabad district

Normalize case, spaces and punctuation. Try matching state + district first.

If the state match fails, use a district name unique across the four selected states. Record MATCH_STATUS = historical_state_crosswalk.

Common project ID: IND_ANDHRAPRADESH_ADILABAD

Year and season link crop records to weather

Final join key: (IDREGION, FYEAR, SEASON)

Season for crop year Y	Weather window
Kharif	June–November of Y
Rabi	November–December of Y−1 and January–April of Y
Summer	March–June of Y
Autumn	September–December of Y
Winter	December of Y−1 and January–March of Y

Adilabad / 2001 / Kharif receives June–November 2001 weather. Summarize many daily rows into one seasonal row. These windows are fixed project assumptions.

Daily weather collapses into one seasonal row

Daily records for the same district	Season window
Day 1: temperature, rain, humidity and other channels	June–November 2001
Day 2: temperature, rain, humidity and other channels	June–November 2001
… remaining complete weather days …	June–November 2001

ONE RECORD

Adilabad
2001
Kharif

SEASONAL FEATURES

Mean temperature, humidity, wind and pressure. Total rainfall, evaporation and radiation. Count complete days and accumulate GDD.

Calculated features add crop and weather context

Engineered feature	Calculation
AREA_FRACTION	District rice area / state rice area for the same year and season
CROP_AREA_LOG1P	log(1 + CROP_AREA)
GDD_BASE10	Sum of max(daily TAVG − 10°C, 0)
TEMP_RANGE_MEAN	TMAX_MEAN − TMIN_MEAN
WATER_BALANCE_PROXY	PREC_TOTAL − ET0_TOTAL
RAIN_PER_GDD	PREC_TOTAL / GDD_BASE10
RADIATION_PER_GDD	RAD_TOTAL / GDD_BASE10

Set the two GDD ratios to zero when GDD is zero. Water balance is a proxy and excludes irrigation.

Merged dataset structure

3,554 rice labels become 2,018 aligned observations

Column role	Count	Contents
Numerical predictor candidates	21	Crop area, location, weather and derived features
Categorical predictor candidates	2	STATE, SEASON
Prediction target	1	YIELD in tonnes per hectare
Identity and audit fields	6	CROP, IDREGION, DISTRICT, FYEAR, MATCH_STATUS, PRODUCTION_TONNES

rice_master.csv contains 30 columns. Retain matched locations, finite yield and at least 60 complete weather days. No simulated observations are added.

Chronological split before fitted preprocessing

Partition	Years	Rows	Use
Training	2001–2015	1,485	Fit feature transformations
Validation	2016–2017	245	Select the DAE checkpoint
Test	2018–2019	288	Preserve a held-out partition

Fit medians, scaling statistics and category mappings on training rows only.

Reuse those fitted transformations for validation and test. Keep identical rows and labels in both representations.

One prepared dataset branches into two representations

MERGED INPUTS

21 numerical fields
STATE and SEASON

TRAIN-FITTED TRANSFORMS

Median imputation and numerical scaling
One-hot encoding for state and season

STANDARD

Keep 21 scaled values
Add 7 indicators

DAE

Learn 8 latent values
Add 7 indicators + score

Both branches keep the same observations, YIELD labels and chronological partitions.

Standard preprocessing preserves named features

21 NUMERICAL FIELDS
Median imputation
Training mean / SD scaling

→

21 SCALED VALUES

Original feature meanings
New comparable scales

2 CATEGORIES

STATE and SEASON
7 one-hot indicators

Standard representation: 21 + 7 = 28 predictors

Adilabad, Kharif 2001	Before scaling	After scaling
CROP_AREA	67,358 hectares	0.2046
TAVG_MEAN	25.8498°C	−0.1383
STATE / SEASON	Andhra Pradesh / Kharif	Corresponding indicators = 1

DAE preprocessing learns eight numerical features

Only the 21 standardized numerical fields enter the autoencoder



Encoder: learn a compact representation

Decoder: reconstruct clean inputs

During training, add Gaussian noise (SD 0.05) to inputs. Keep clean values as reconstruction targets. Export latent features using clean inputs.

8 latent features + 7 category indicators + 1 reconstruction MSE = 16 predictors

Category names become numerical indicators

Example: Andhra Pradesh, Kharif

State indicators	AP	Karnataka	Tamil Nadu	Telangana
Encoded values	1	0	0	0

Season indicators	Kharif	Rabi	Summer
Encoded values	1	0	0

Four state columns + three training-season columns = seven indicators. Both representations use these same columns.

Standard representation: 21 scaled numerical features

numeric__CROP_AREA
numeric__CENTROID_X
numeric__CENTROID_Y
numeric__WEATHER_DAY_COUNT
numeric__TMAX_MEAN
numeric__TMIN_MEAN
numeric__TAVG_MEAN
numeric__VPRES_MEAN
numeric__WSPD_MEAN
numeric__PREC_TOTAL
numeric__ET0_TOTAL
numeric__RAD_TOTAL
numeric__RELH_MEAN
numeric__PS_MEAN
numeric__GDD_BASE10
numeric__AREA_FRACTION
numeric__TEMP_RANGE_MEAN
numeric__WATER_BALANCE_PROXY
numeric__RAIN_PER_GDD
numeric__RADIATION_PER_GDD
numeric__CROP_AREA_LOG1P

The numeric__ prefix identifies the scaled output. Each column keeps its original feature meaning.

Both versions share seven categorical indicators

categorical__STATE_Andhra Pradesh

categorical__STATE_Karnataka

categorical__STATE_Tamil Nadu

categorical__STATE_Telangana

categorical__SEASON_Kharif

categorical__SEASON_Rabi

categorical__SEASON_Summer

Values are 0 or 1. Training contains four states and three seasons. Autumn and Winter are unseen training seasons, so their season indicators become all zeros in held-out records.

DAE representation: all 16 predictor columns

8 learned numerical features

AE_LATENT_01

AE_LATENT_02

AE_LATENT_03

AE_LATENT_04

AE_LATENT_05

AE_LATENT_06

AE_LATENT_07

AE_LATENT_08

7 unchanged category indicators

categorical__STATE_Andhra Pradesh

categorical__STATE_Karnataka

categorical__STATE_Tamil Nadu

categorical__STATE_Telangana

categorical__SEASON_Kharif

categorical__SEASON_Rabi

categorical__SEASON_Summer

AE_RECONSTRUCTION_MSE

The original 21 numerical columns are replaced. YIELD and metadata remain separate from predictors.

Why we created two representations

Difference	Standard	DAE
Numerical information	21 scaled original features	8 learned combinations
Interpretation	Named physical measurements	Latent values have no fixed physical label
Total predictor count	28	16
Additional feature	None	Reconstruction MSE
Rows, labels and time splits	Identical	Identical

Standard provides the conventional reference. DAE tests whether compact learned features help. Compression can discard useful information, so fewer columns do not guarantee better prediction.

The same observation has two preprocessed forms

Adilabad, Kharif 2001	Standard representation	DAE representation
Numerical values	21 scaled named features	AE_LATENT_01 … AE_LATENT_08
Example numerical values	CROP_AREA: 0.2046 TAVG_MEAN: −0.1383	LATENT_01: −0.9492 LATENT_02: −1.5666
State and season	7 one-hot indicators	Same 7 one-hot indicators
Additional score	None	AE_RECONSTRUCTION_MSE: 0.01388
Target, kept separate	YIELD: 2.231999 t/ha	Same YIELD: 2.231999 t/ha

Latent values combine input patterns. They do not individually mean rainfall or temperature.

Training-ready dataset outputs

Output files	Rows per file	Predictors	Stored columns
standard/train.csv	1,485	28	34
standard/validation.csv	245	28	34
standard/test.csv	288	28	34
autoencoder/train.csv	1,485	16	22
autoencoder/validation.csv	245	16	22
autoencoder/test.csv	288	16	22

Each CSV also stores YIELD and five metadata fields. manifest.json lists the exact predictor columns. Select inputs explicitly and keep the partitions separate.

Each representation exports three separate CSVs

STANDARD

28 predictor columns
YIELD + 5 metadata

→

STANDARD OUTPUTS

train.csv: 1,485 rows
validation.csv: 245 rows
test.csv: 288 rows

DAE

16 predictor columns
YIELD + 5 metadata

→

DAE OUTPUTS

train.csv: 1,485 rows
validation.csv: 245 rows
test.csv: 288 rows

Six training-ready CSVs. The manifest specifies which columns become inputs and which remain context.